



PHILADELPHIA
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Rooted in Brotherly Love & Lifelong Community

ANNUAL
MEETING

Encephalopathy of Prematurity: What Clinicians Need to Know

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**Research
Saves Lives**





Speaker Name: Lauren Leigh Cooney Jantzie, PD

1. Disclosure of Relevant Financial Relationships

I have no financial relationships to disclose.

2. Disclosure of Off-Label and/or investigative uses:

I will discuss the following off label use and/or investigational use in my presentation:

- IND approved, off label use of erythropoietin (EPO) and melatonin (MLT)
- *Safety of combined therapy with erythropoietin and melatonin for preterm infants with intraventricular hemorrhage (SCEMPI)*
 - NCT05617833
 - IRB00301237
 - 1R01HD104673-01A1
 - PCT/US2022/018046

**Research
Saves Children**



CP: Everything, Everywhere, All at Once

The pathogenesis that leads to cerebral palsy is multifactorial and diverse

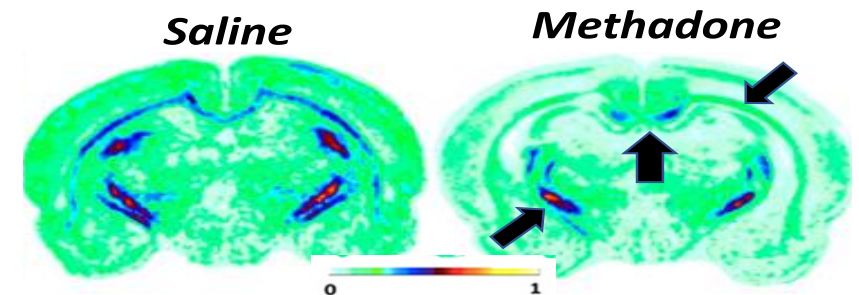
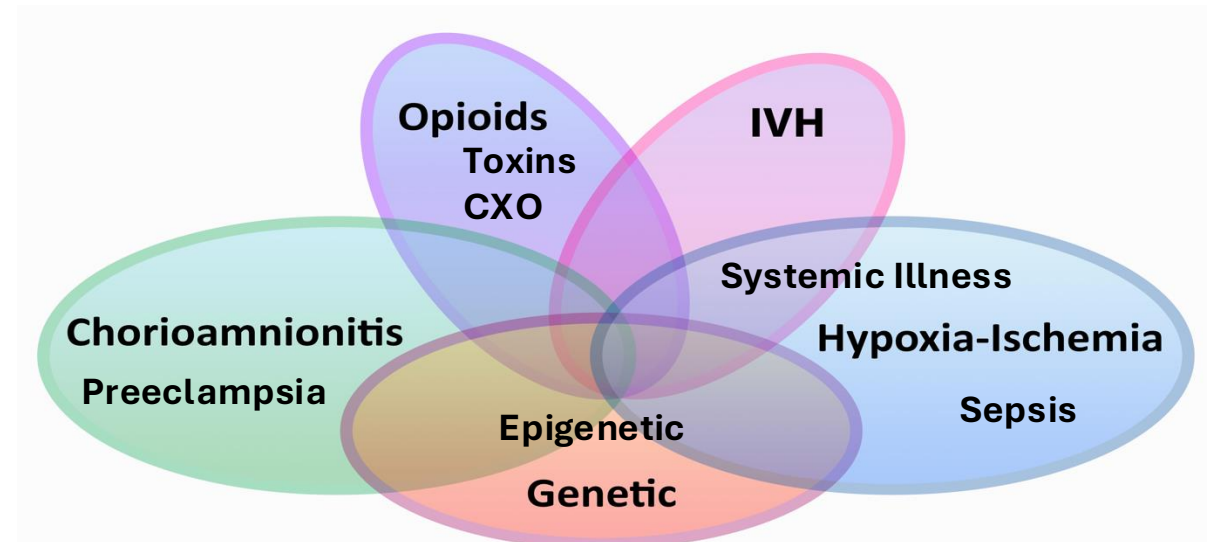
- Prematurity
 - Chorioamnionitis
 - IVH

❖ Genetics

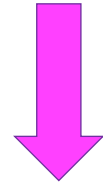
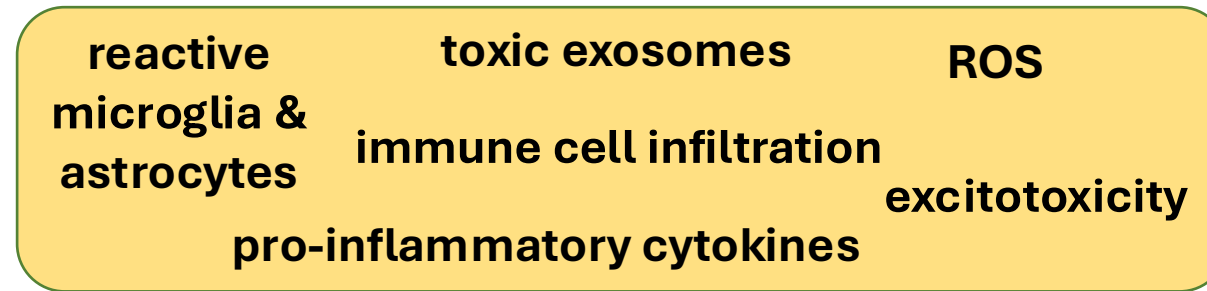
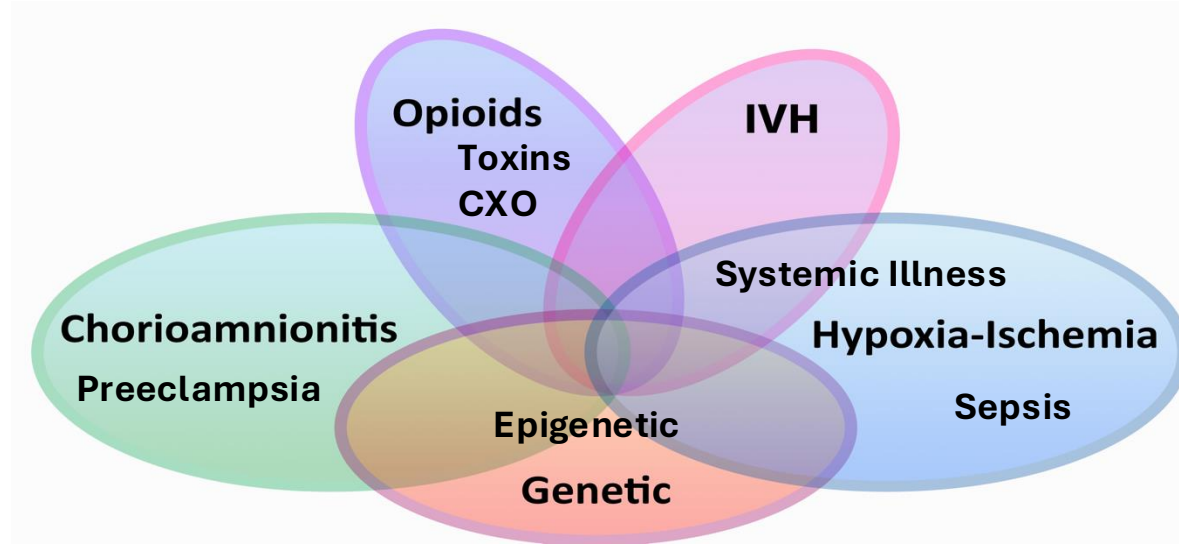
❖ Stroke/Trauma

❖ Opioids

- **Methadone**, morphine, fentanyl (prenatally and postnatally)
- NOWS
- Pain
- Polysubstance use



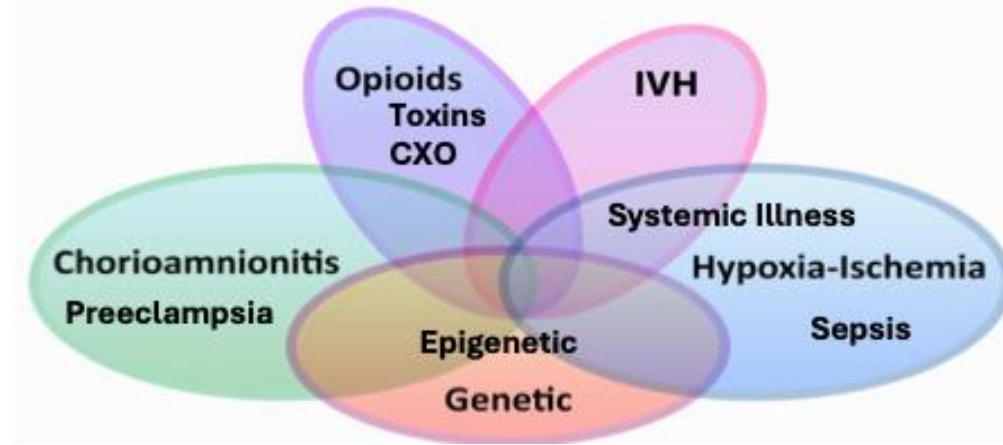
Embracing the Diversity of Brain Injury



Damage to Developing CNS

****cognitive, executive, motor & neuropsychological outcomes****

Consequences are Diverse



White Matter Injury

Gait & Motor Deficits

Cognitive Deficits & Executive Dysfunction

Hydrocephalus

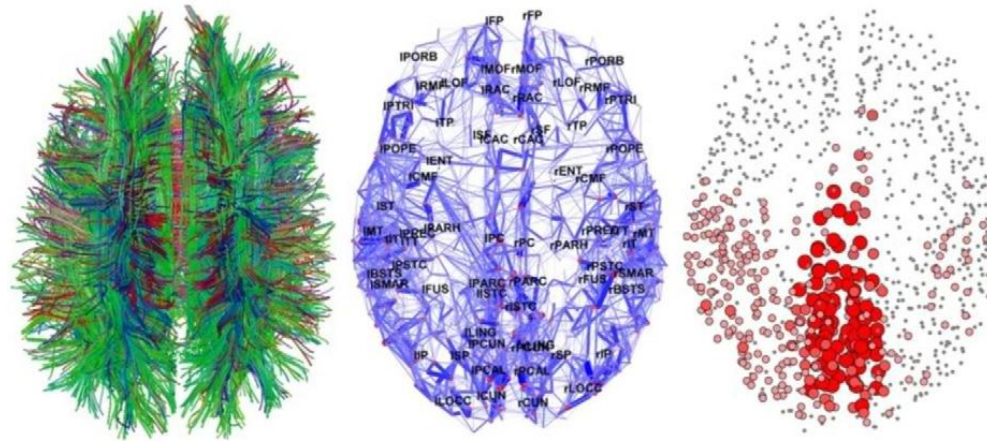
Acute and Chronic Pain

Effects of Impaired Neurodevelopment Through Adulthood

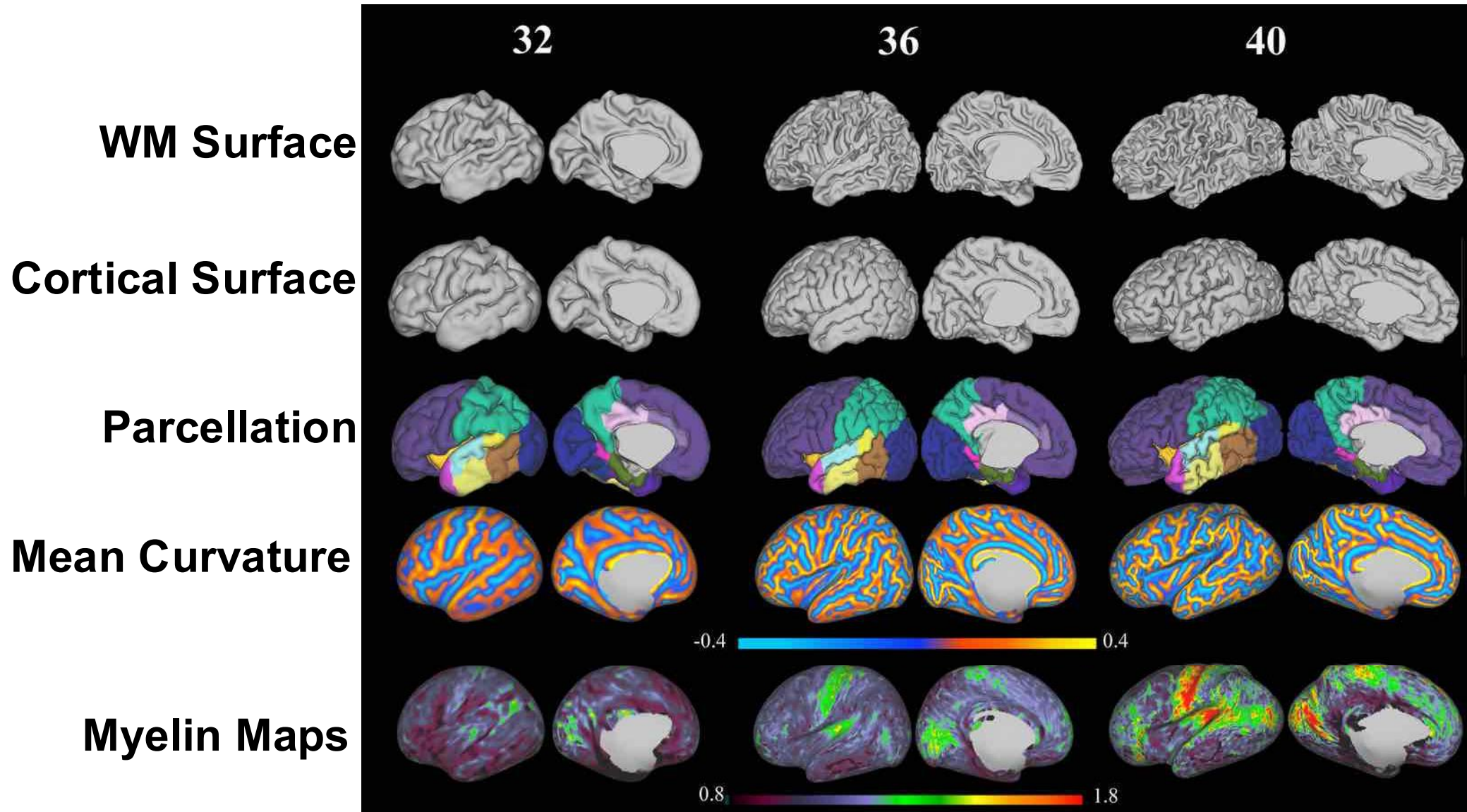
It's Complicated

(another movie title)

- The precise mechanisms linking perinatal brain injury to long-term structural and functional abnormalities are unknown
 - Neural network fragmentation
 - Impaired anatomical connectivity
 - Diminished elaboration of functional connectomes

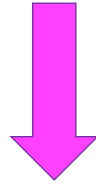


An Elaborate Developmental Program



Inherent Issues with Neural Networks

- **Fragmented functional networks**
 - ❖ (less integration)
- **Reduced intra-hemispheric connectivity**
 - ❖ (less transmission capacity)
- **Increased spectral energy but reduced connectivity**
 - ❖ (less global efficiency)
- **Abnormal lateralization**
 - ❖ (Less specialization/local information processing)



****cognitive, executive, motor & neuropsychological outcomes****

Edwards et al., 2017

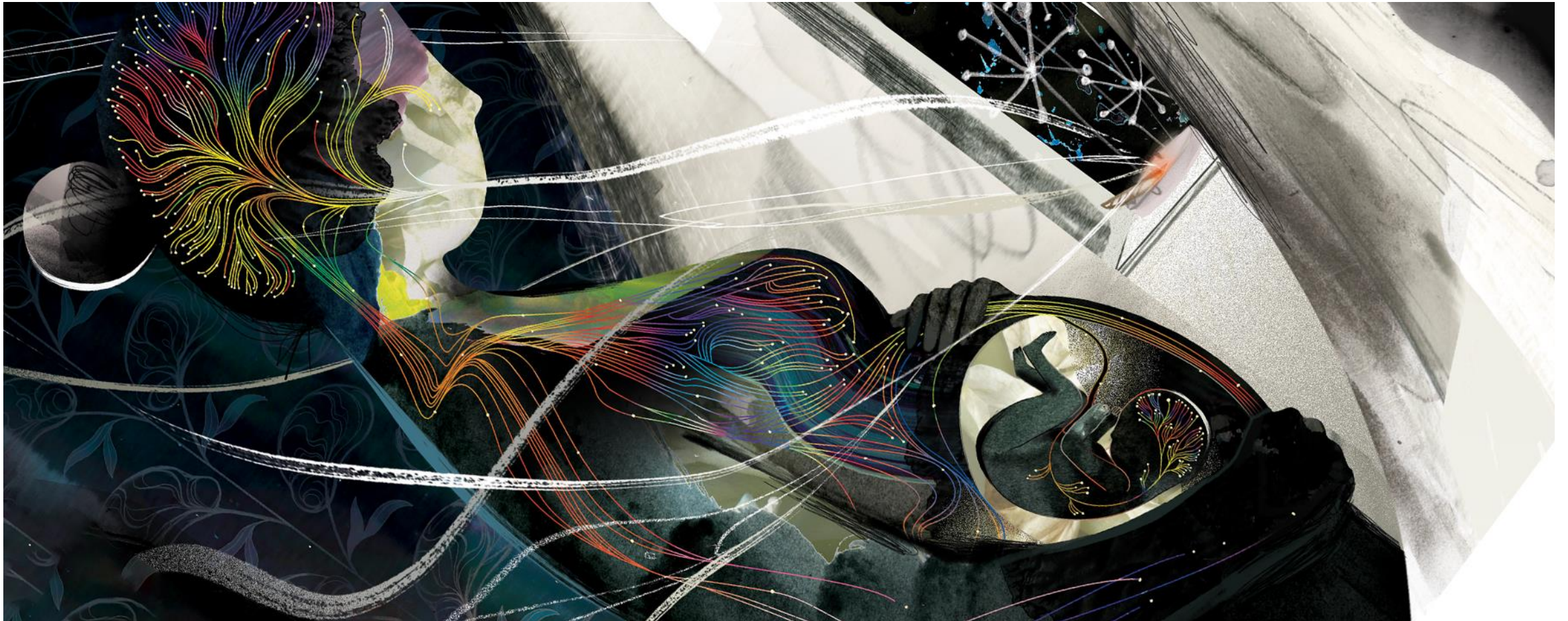
Schienost 2016

Thomason et al., 2017

Ment et al., 2017

But Why?

Dysfunction in Placental-Fetal-Brain Axis

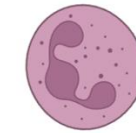


But Why?

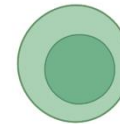
Systemic Immune- Neural Immune Communication

Inflammatory Signal Transduction

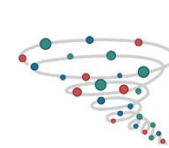
- ✓ Immune activation
- ✓ Inflammatory priming
- ✓ Dynamic population shifts
- ✓ Immune memory
- ✓ Developmental trajectory changes



Neutrophil



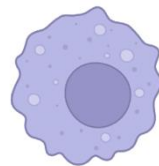
T cell



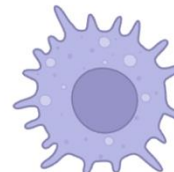
Cytokine storm



Toll-like
receptor



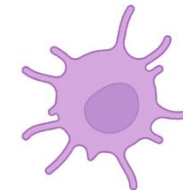
Resting
macrophage



Activated
macrophage



Basophil



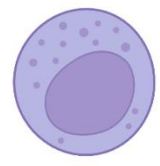
Dendritic cell



Eosinophil



Monocyte



Natural Killer Cell

The Duality of Inflammation



DC



NK cell



T_{CM} cell



T_{EM} cell



Microglia



BAM



MDM



Monocyte



γδ T cell



T_{reg} cell



IL-10



IL-4



IFNγ



Neutrophil



Mast cell

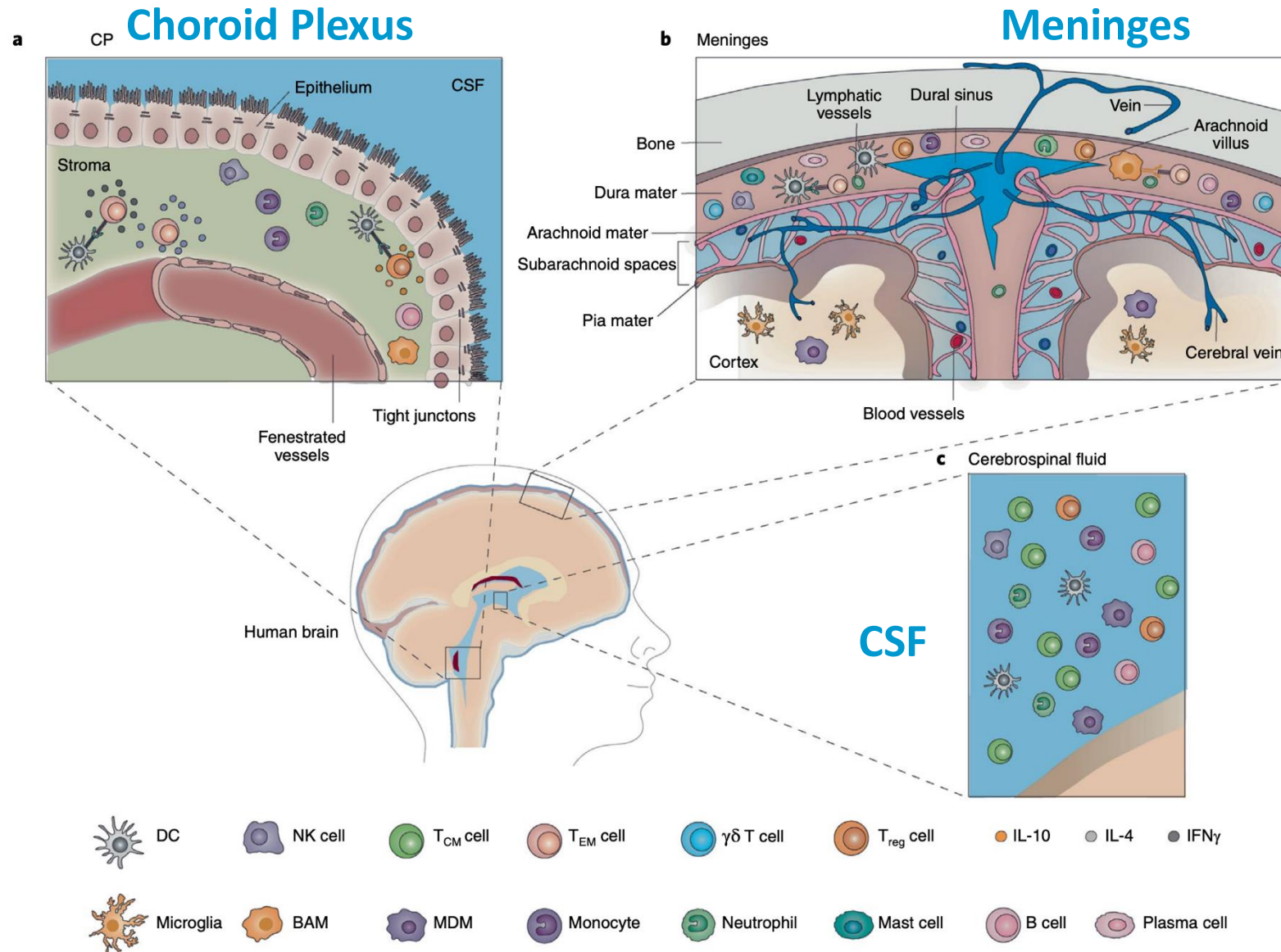


B cell



Plasma cell

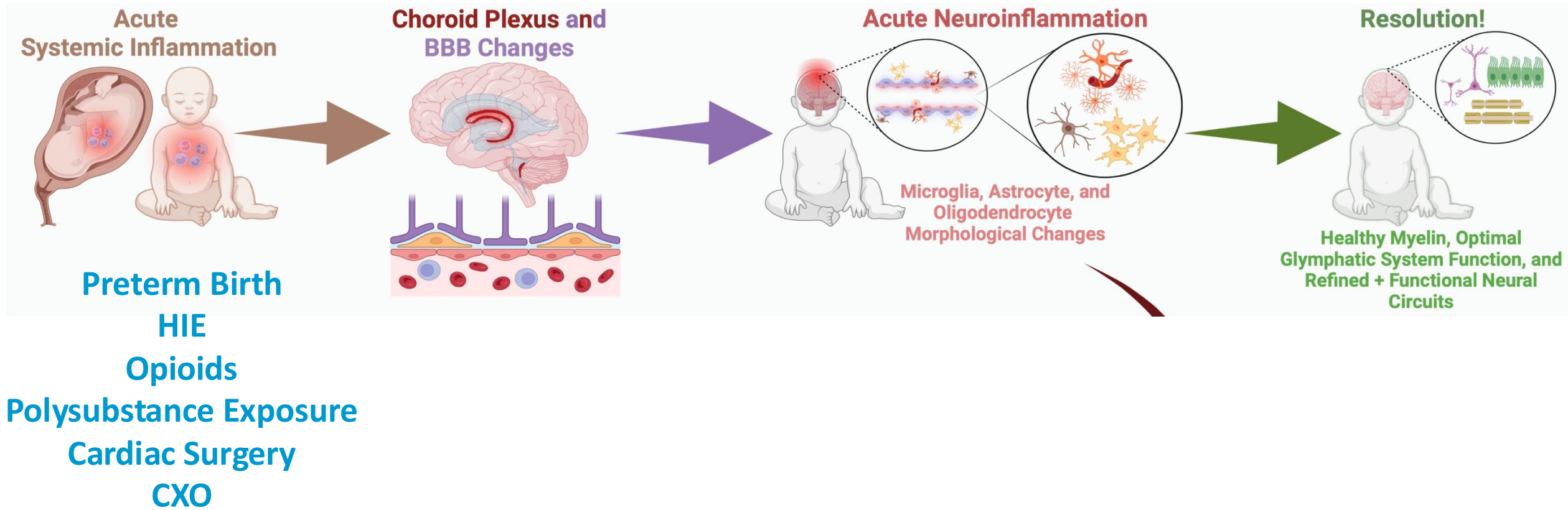
Professional Immune Cells



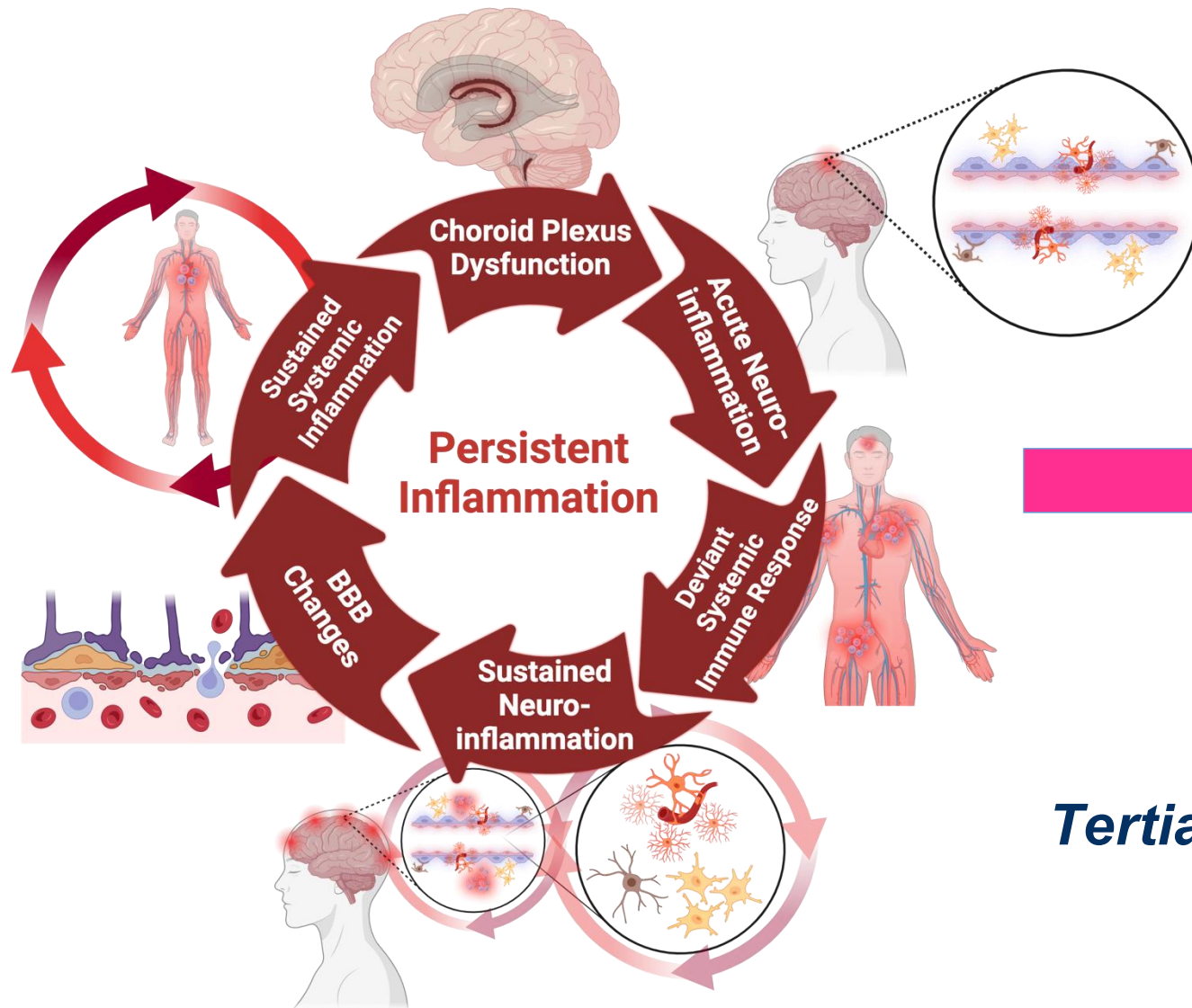
Pathogenic Immunity

- ✓ Persistent, sustained, hyper-reactive peripheral immune cells
 - ✓ Educated immune cells in brain and body (i.e. blood, gut, meninges, Cpx)
 - ✓ Toxic cytokines & metabolic dysfunction
 - ✓ Dysregulated Tcells, Neutrophils and macrophages
 - ✓ Neuroinflammation
-
- ✓ Fragmented neural networks
 - ✓ Impaired connectivity
 - ✓ Decreased cognitive and motor performance
 - ✓ Chronic pain

Perinatal Brain Injury: Ideal Scenario



The Reality of the Pathophysiology



- ✓ Dynamic population shifts
- ✓ Immune memory
- ✓ Toxic Cpx
- ✓ Disruptions in glymphatic clearance
- ✓ Fragmented neural networks
- ✓ PNN degradation
- ✓ Disrupted anatomical and functional connectivity

Tertiary Brain Damage: Failed Recovery

What is Different in People with CP?

Immune Cell Priming from Chronic Systemic Inflammation

Altered Inflammatory Responses in Preterm Children with Cerebral Palsy

Chang-Yi Lin, BS,¹ Ying-Chao Chang, MD, PhD,² Shan-Tair Wang, PhD,³
Ting-Yang Lee, MS,⁴ Chiou-Feng Lin, PhD,⁵
and Chao-Ching Huang, MD^{5,6}

Cytokine dysregulation in children with cerebral palsy

ZUNERA ZAREEN^{1,2,3,4}  | TAMMY STRICKLAND^{1,2} | LIDA FALLAH^{2,5} | VICTORIA MCENEANEY^{1,2} |
LYNNE KELLY^{1,2,6} | DENISE MCDONALD^{1,3} | ELEANOR J MOLLOY^{1,2,3,4,6,7} 

CLINICAL RESEARCH ARTICLE **OPEN**



Severe neurological impairment and immune function: altered neutrophils, monocytes, T lymphocytes, and inflammasome activation

John Allen^{1,2,3,4}, Johana Isaza-Correa^{1,2,3}, Lynne Kelly^{1,2,3}, Ashanty Melo^{1,2,3}, Aoife Mahony⁴, Denise McDonald^{1,2,4} and Eleanor J. Molloy^{1,2,3,4,5,6} 

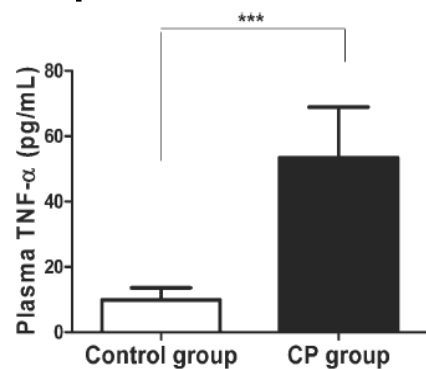
Sustained Peripheral Immune Hyper-Reactivity (SPIHR):

The Immune Landscape Captures the Complexity of Disease in CP

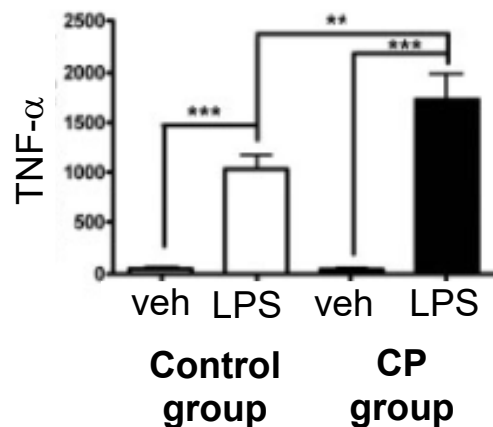
Compare peripheral blood of 5-year-old children who were all born preterm:

Cerebral Palsy Versus No Cerebral Palsy

plasma $\text{TNF-}\alpha$

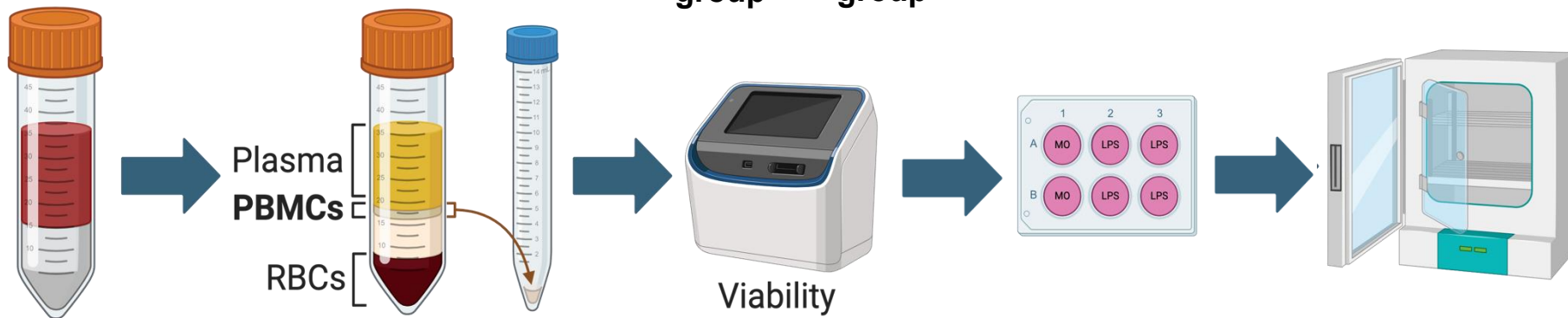


$\text{TNF-}\alpha$



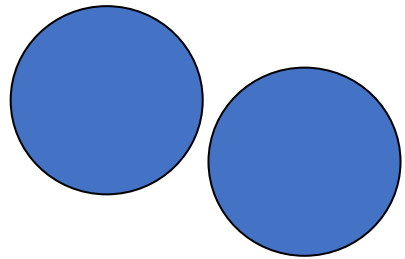
PBMC
isolation

- ✓ T Cells
- ✓ T Regs
- ✓ T Helper
- ✓ B Cells
- ✓ Monocytes
- ✓ NK
- ✓ Dendritic cells



Sustained Peripheral Immune Hyper-Reactivity (SPIHR):

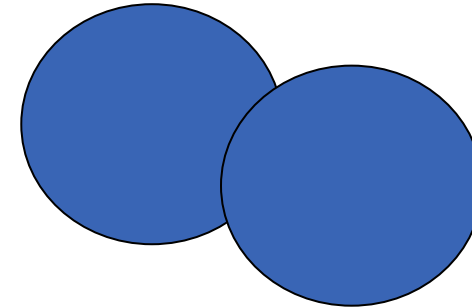
Lymphocytes



*Infection
Challenge*

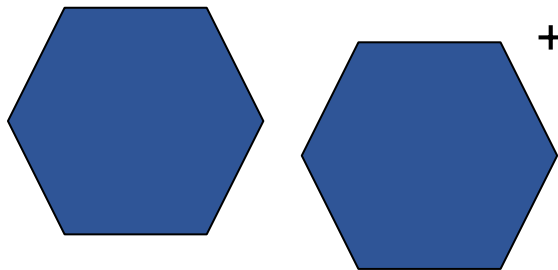


Activated



+ cytokines
i.e. TNF α

*Sensitized or Primed Lymphocyte
(Former Preterm Infant with CP)*

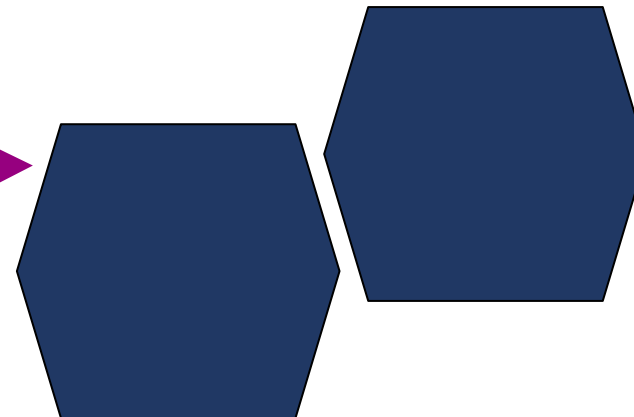


+++memory

*Infection
Challenge*

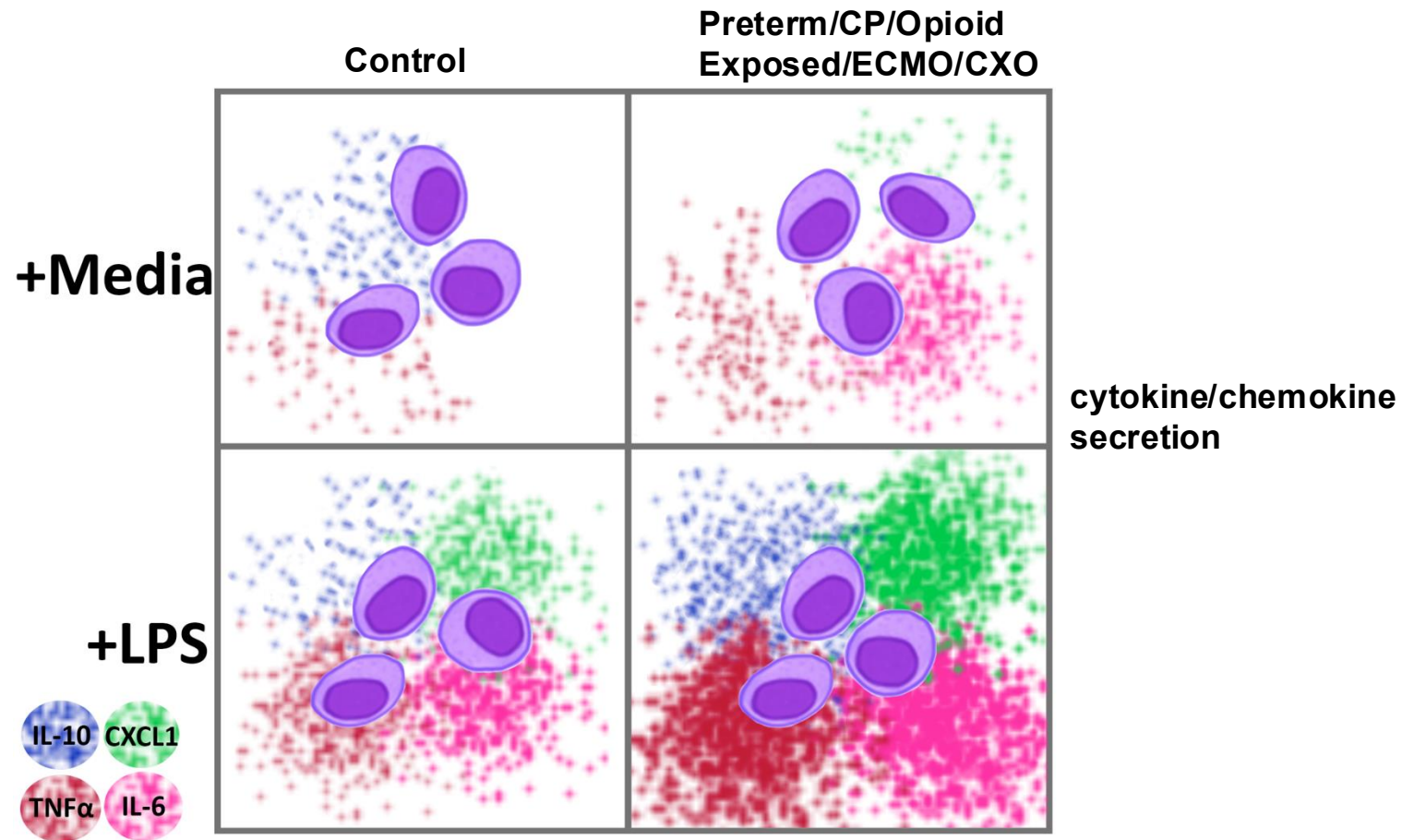


Hyper-Activated

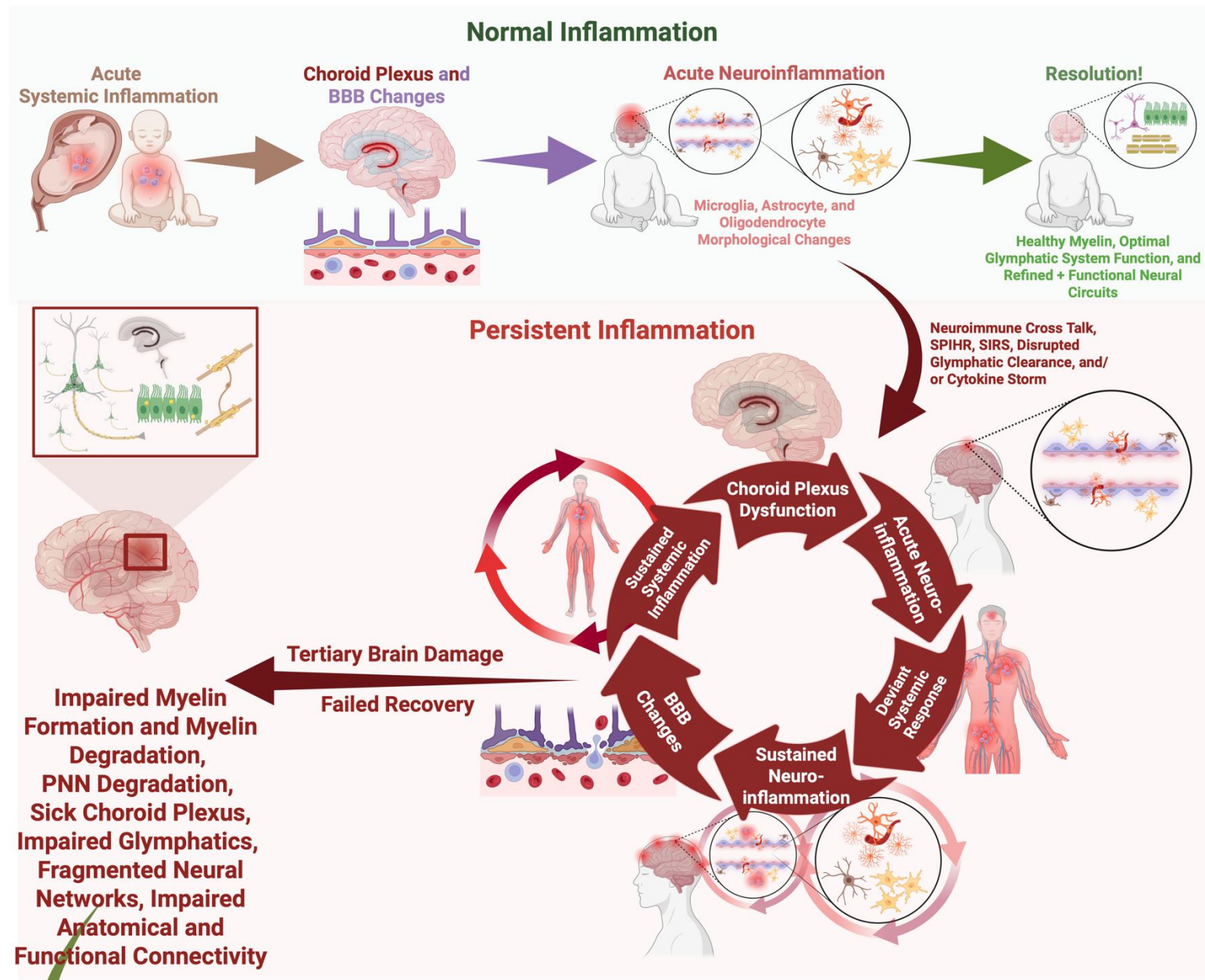


+++ TNF α
+++duration of
activation
+++

Sustained Peripheral Immune Hyper-Reactivity (SPIHR):



The Pathophysiology is Complex



The Pathophysiology is Complex

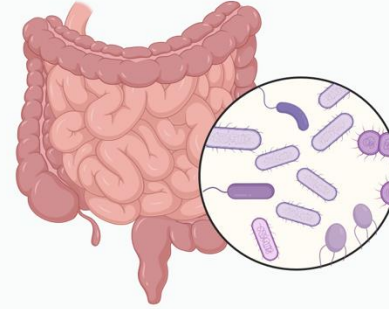
Treatment Options After the Neonatal Period



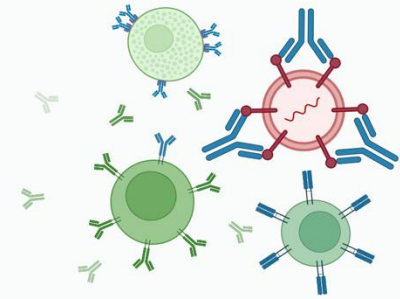
Caffeine



Melatonin



Microbiome Optimization/
Probiotics



Novel Immunomodulation

Conclusions

- perinatal brain injury and its LONG-TERM pathophysiology
 - **Persistent Inflammation and FAILED RECOVERY**
- SPIHR is a unique signature across spectrum of perinatal and developmental brain injuries
 - Chorioamnionitis
 - Opioids
 - HI
 - NE
 - CP
 - Trisomy 21
 - ECMO
 - THC
 - Cardiac Surgery, CXO
- Immunomodulatory therapies that repair disordered brain and immune system developmental trajectory and function may be beneficial therapeutic approaches